Discrete Random Variables
Reading Objectives:

In this reading, you will be introduced to the probability density function of a discrete random variable and its mean, variance, and standard deviation with examples.

Main Reading Section:

Definition 1: The cumulative distribution function, or more simply the **distribution function**, F of the random variable X is defined for any real number x by

$$F(x) = P\{X \leq x\}$$

That is, $F(x)$ is the probability that the random variable X takes on a value that is less than or equal to x.

Example 1: Suppose X has a probability mass function given $p(1) = \frac{1}{2}, p(2) = \frac{1}{3}, p(3) = \frac{1}{6}$

Then, the cumulative distribution function F of X is given by

$$F(a) = \begin{cases} 0, & a < 1 \\ \frac{1}{2}, & 1 \leq a < 2 \\ \frac{5}{6} & 2 \leq a < 3 \\ 1 & 3 \leq a \end{cases}$$

Expectation (Mean)

Definition 2: If X is a discrete random variable taking on the possible values $x_1, x_2 ...$, then the expectation or expected value of X, denoted by $E[X]$, is defined by

$$E[X] = \sum_i x_i \, P\{X = x_i\}$$

Proposition 1: (Expectation of a function of a random variable).

If X is a discrete random variable with probability mass function $p(x)$, then for any real-valued function g

$$E[g(X)] = \sum_x g(x)p(x)$$

Variance and Standard deviation

Definition 3: If X is a random variable with mean μ, then the variance of X, denoted by $\text{Var}(X)$, is defined by

$$\text{Var}(X) = E[(X - \mu)^2] \text{ or } \text{Var}(X) = E[X^2] - E[X]^2$$

Definition 4: The quantity $\sqrt{\text{Var}(X)}$ is called the standard deviation of X.

Remarks:

$$\text{Var}(aX + b) = a^2 \ \text{Var}(X)$$
$$\text{Var}(aX) = a^2 \ \text{Var}(X)$$

Example 2: Compute $E(X)$ when X represents the outcome when we roll a fair die.

Solution: Since $p(1) = p(2) = p(3) = p(4) = p(5) = p(6) = \frac{1}{6}$, we obtain that

$$E[X] = 1(\tfrac{1}{6}) + 2(\tfrac{1}{6}) + 3(\tfrac{1}{6}) + 4(\tfrac{1}{6}) + 5(\tfrac{1}{6}) + 6(\tfrac{1}{6}) = \tfrac{7}{2}$$

Example 3: Compute $\text{Var}(X)$ when X represents the outcome when we roll a fair die.

Solution: Since $P\{X = i\} = \frac{1}{6}$ where $i = 1, 2, 3, 4, 5, 6$, we obtain

$$E[X^2] = \sum_{i=1}^{6} i^2 P\{X = i\}$$
$$= 1^2\left(\frac{1}{6}\right) + 2^2\left(\frac{1}{6}\right) + 3^2\left(\frac{1}{6}\right) + 4^2\left(\frac{1}{6}\right) + 5^2\left(\frac{1}{6}\right) + 6^2\left(\frac{1}{6}\right)$$
$$= \frac{91}{6}$$

Through the previous example $E[X] = \frac{7}{2}$. Hence, we obtain

Var(X)= $E[X^2] - E[X]^2 = \frac{91}{6} - (\frac{7}{2})^2 = \frac{35}{12}$